

Mixtures Aggregating Genes Into Coordinated Modules And Pathways (MAGICMAP)

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1 Introduction

We present a new tool, Mixtures Aggregating Genes Into Coordinated Modules And Pathways (MAGICMAP), to identify mechanistic patterns in genetic data. We first describe a motivating theoretical model, followed by the statistical model we propose fitting to results from genetic association studies.

2 Theoretical model

2.1 Framework

Let phenotypes, P_1 and P_2 , be composed of mechanisms M_i , and let mechanisms be affected by genetic variants g_j . Then we can suppose a simple linear model

$$P_1 = \sum_{i \in T_1} \gamma_i M_i + \varepsilon_1 \quad (1)$$

$$P_2 = \sum_{i \in T_2} \delta_i M_i + \varepsilon_2 \quad (2)$$

$$M_i = \sum_{j \in S_i} \alpha_{i,j} g_j + \varepsilon_{M_i} \quad (3)$$

where T_1 , T_2 , and S_i denote the set of mechanisms or variants that have non-zero effect sizes γ_i , δ_i , or $\alpha_{i,j}$, respectively. It can be shown that this general model is consistent with a broad set of possible causal models for the relationship between the two phenotypes and genetic variation. Collapsing across mechanisms, this implies a conventional linear model for the effects of genetic variants on phenotypes.

$$P_1 = \sum_{i \in T_1} \gamma_i \sum_{j \in S_i} \alpha_{i,j} g_j + (\gamma_i \varepsilon_{M_i} + \varepsilon_1) = \sum_{i \in T_1} \sum_{j \in S_i} \gamma_i \alpha_{i,j} g_j + e_1 \quad (4)$$

$$P_2 = \sum_{i \in T_2} \delta_i \sum_{j \in S_i} \alpha_{i,j} g_j + (\delta_i \varepsilon_{M_i} + \varepsilon_2) = \sum_{i \in T_2} \sum_{j \in S_i} \delta_i \alpha_{i,j} g_j + e_2 \quad (5)$$

From genome-wide association studies (GWAS), we observe estimates of the marginal association of a genetic variant with the phenotype. For a given SNP x_j , the expected values of this association are

$$\beta_{1,j} = \frac{Cov(x_j, P_1)}{Var(x_j)} \quad (6)$$

$$\beta_{2,j} = \frac{Cov(x_j, P_2)}{Var(x_j)} \quad (7)$$

where in practice all terms here will be considered after residualizing on covariates.

2.2 Comparing effect sizes

To think about the relationship between these two effect sizes, we consider the ratio

$$\theta_j = \frac{\beta_{2,j}}{\beta_{1,j}} = \frac{Cov(x_j, P_2)/Var(x_j)}{Cov(x_j, P_1)/Var(x_j)} = \frac{Cov(x_j, P_2)}{Cov(x_j, P_1)} \quad (8)$$

This ratio is defined as long as the variant has a true relationship to the first phenotype ($\beta_{1,j} \neq 0$). We assume here that we are considering the two effect sizes in the same population with functionally equivalent covariates, such that the variance of the SNP is the same in the numerator and denominator allowing the last simplification.

It follows then that

$$\theta_j = \frac{Cov(x_j, P_2)}{Cov(x_j, P_1)} = \frac{Cov(x_j, \sum_{i \in T_2} \sum_{k \in S_i} \delta_i \alpha_{i,k} g_k + e_2)}{Cov(x_j, \sum_{i \in T_1} \sum_{k \in S_i} \gamma_i \alpha_{i,k} g_k + e_1)} \quad (9)$$

$$\theta_j = \frac{\sum_{i \in T_2} \sum_{k \in S_i} \delta_i \alpha_{i,k} Cov(x_j, g_k) + Cov(x_j, e_2)}{\sum_{i \in T_1} \sum_{k \in S_i} \gamma_i \alpha_{i,k} Cov(x_j, g_k) + Cov(x_j, e_1)} \quad (10)$$

It can be shown that under specific conditions this ratio will be the same for multiple SNPs x_j that have the same mechanistic relationship to the two phenotypes. For simplicity, consider SNPs that are not related to residual environmental effects, $Cov(x_j, e_1) = Cov(x_j, e_2) = 0$. In that case, SNPs that only tag effects for a single mechanism M_* will have the same expected θ_j .

More formally, this is the set of variants S_* where

$$Cov(x_j, g_k) = 0, \quad \forall k \in \bigcup_{i \neq *} S_i \quad (11)$$

Note that this condition isn't simply that $Cov(x_j, g_k) = 0$ with variants g_k that aren't in S_* . The theoretical model allows that some variants g_k might be members of both the set S_* related to mechanism M_* and one or more other sets S_i for other mechanisms. The requirement here is that $Cov(x_j, g_k) = 0$ if g_k is in one of those other sets regardless of being in set S_* .

For SNPs x_j in S_* , we can observe that the terms for other S_i disappear leaving

$$\theta_j = \frac{\sum_{k \in S_*} \delta_* \alpha_{*,k} \text{Cov}(x_j, g_k)}{\sum_{k' \in S_*} \gamma_* \alpha_{*,k'} \text{Cov}(x_j, g_{k'})} \quad (12)$$

$$\theta_j = \frac{\delta_* \sum_{k \in S_*} \alpha_{*,k} \text{Cov}(x_j, g_k)}{\gamma_* \sum_{k \in S_*} \alpha_{*,k} \text{Cov}(x_j, g_k)} \quad (13)$$

$$\theta_j = \frac{\delta_*}{\gamma_*} \quad (14)$$

Notably this last step assumes $\text{Cov}(x_j, g_k)$ is the same in both GWAS, which should hold as long as the two GWAS were performed in the same genetic population (i.e. with the same allele frequencies and patterns of linkage disequilibrium), and that the effects $\alpha_{*,k}$ are the same in both GWAS populations (e.g. that they do not differ due to variation from GxE or GxG effects with differential background in the two cohorts). In this scenario, all SNPs j meeting this criteria for the same set S_* for mechanism M_* will have the same θ_j ratio.

Because this set of SNPs S_* all relate to only this one mechanism M_* without tagging any other mechanisms we interpret these SNPs as core to mechanism M_* , making them distinct from any other SNPs that are related to mechanism M_* but also have pleiotropic relationship to additional M_* .

It can be shown that a constant ratio θ_j for multiple SNPs x_j can also be observed under other conditions, but in each case they reflect a set of variants that share the same structure of how they relate to the pair of phenotypes being studied. Thus in all scenarios where we observe a set of SNPs with the same ratio θ_j we posit that they are potentially the core of some mechanism M_* .

3 MAGICMAP regression mixture model

3.1 Model for observed GWAS effect sizes

Let $\hat{\beta}_{1,j}$ and $\hat{\beta}_{2,j}$ be estimated marginal effects for SNP j from GWAS of phenotypes 1 and 2, respectively. In practice, we select SNPs j that are independent lead genome-wide significant SNPs from GWAS of phenotype 1. We assume that these effects are measured with sampling error relative to the true population effect sizes

$$\hat{\beta}_{1,j} = \beta_{1,j} + e_{1,j} \quad (15)$$

$$\hat{\beta}_{2,j} = \beta_{2,j} + e_{2,j} \quad (16)$$

$$\begin{bmatrix} e_{1,j} \\ e_{2,j} \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{1,j}^2 & \rho \sigma_{1,j} \sigma_{2,j} \\ \rho \sigma_{1,j} \sigma_{2,j} & \sigma_{2,j}^2 \end{bmatrix} \right) \quad (17)$$

It is assumed that the GWAS results are from a conventional regression model (e.g. linear or logistic regression, as appropriate) with large enough sample sizes for the sampling error to be normally distributed (i.e. following the central limit theorem). The sampling error is allowed to be

correlated to reflect potential sample overlap between the two GWAS, with the correlation ρ assumed to be the same across all SNPs j .

From the theoretical model, we expect that SNPs from a genetic component k with a given distinct relationship to the two phenotypes will have the same proportional relationship between their population effect sizes, $\theta_k = \beta_{2,j}/\beta_{1,j}$, which by rearrangement implies the regression relationship

$$\beta_{2,j} = \theta_k \beta_{1,j} \quad (18)$$

Letting z_j be an indicator for which component SNP j is a member of, we can write the conditional distribution of the observed GWAS estimates as

$$f\left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix} \middle| z_j = k\right) = \varphi\left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix}; \begin{bmatrix} \beta_{1,j} \\ \theta_k \beta_{1,j} \end{bmatrix}, \begin{bmatrix} \sigma_{1,j}^2 & \rho \sigma_{1,j} \sigma_{2,j} \\ \rho \sigma_{1,j} \sigma_{2,j} & \sigma_{2,j}^2 \end{bmatrix}\right) \quad (19)$$

where $\varphi(Y; \mu, \Sigma)$ is the density of a bivariate normal distribution with mean μ and covariance Σ evaluated at observed values Y .

We then assume that the reported standard errors for the GWAS effect sizes ($\hat{\sigma}_{j,1}^2$ and $\hat{\sigma}_{j,2}^2$) are correctly estimated, and that an estimate of the residual correlation ρ can be obtained as detailed below. Substituting those observed values and treating them as fixed gives

$$f\left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix} \middle| z_j = k\right) = \varphi\left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix}; \begin{bmatrix} \beta_{1,j} \\ \theta_k \beta_{1,j} \end{bmatrix}, \begin{bmatrix} \hat{\sigma}_{1,j}^2 & \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} \\ \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} & \hat{\sigma}_{2,j}^2 \end{bmatrix}\right) \quad (20)$$

This corresponds to a regression mixture model where each component following a York regression model with no intercept.¹⁻³ (Such “errors-in-variables” models have arisen independently multiple times in the literature^{4,5}, but we adopt the language of York regression as our introduction to the space.)

$$f\left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix} \middle| z_j = k\right) = \begin{cases} \hat{\beta}_{1,j} = \beta_{1,j} + e_{1,j} \\ \hat{\beta}_{2,j} = \theta_k \beta_{1,j} + e_{2,j} \\ \begin{bmatrix} e_{1,j} \\ e_{2,j} \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \hat{\sigma}_{1,j}^2 & \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} \\ \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} & \hat{\sigma}_{2,j}^2 \end{bmatrix}\right) \end{cases} \quad (21)$$

Since we don't observe which SNPs belong to which component (i.e. z_j is unknown) we can marginalize across uncertainty, introducing a parameter for the prior probability $P(z_j = k) = \pi_k$.

The distribution of the observed GWAS effect size estimates is then

$$f\left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix}\right) = \sum_{k=1}^K \pi_k \varphi\left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix}; \begin{bmatrix} \beta_{1,j} \\ \theta_k \beta_{1,j} \end{bmatrix}, \begin{bmatrix} \hat{\sigma}_{1,j}^2 & \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} \\ \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} & \hat{\sigma}_{2,j}^2 \end{bmatrix}\right) \quad (22)$$

The likelihood function for the unknown parameters θ_k , π_k , and $\beta_{1,j}$ is then given by

$$\mathcal{L}(\theta, \pi, \beta_1 | \hat{\beta}_1, \hat{\beta}_2, \hat{\sigma}_1^2, \hat{\sigma}_2^2, \hat{\rho}) = \prod_{j=1}^J \sum_{k=1}^K \left[\pi_k \varphi\left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix}; \begin{bmatrix} \beta_{1,j} \\ \theta_k \beta_{1,j} \end{bmatrix}, \begin{bmatrix} \hat{\sigma}_{1,j}^2 & \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} \\ \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} & \hat{\sigma}_{2,j}^2 \end{bmatrix}\right) \right] \quad (23)$$

assuming that the J SNPs are independent.

3.2 Maximum likelihood estimation with the EM algorithm

Direct maximization of the marginal likelihood $\mathcal{L}$ is intractable analytically, thus we perform maximization using the Expectation-Maximization (EM) algorithm. The EM algorithm indirectly maximizes $\mathcal{L}$ by iteratively taking the expectation of the log of the complete data likelihood (i.e. if component memberships z_j were observed) and then maximizing the expected log likelihood to get parameter estimates. It is well established that this approach enables maximum likelihood estimation of mixture models such as the one desired here.^{6,7}

3.2.1 Initial parameter estimates

The EM algorithm requires initial estimates of the parameter values. To get these starting values, we compute the ratio of the observed effect sizes for each variant

$$\hat{\theta}_j = \frac{\hat{\beta}_{2,j}}{\hat{\beta}_{1,j}}$$

and then perform weighted univariate clustering with K clusters as implemented in Ckmeans.1d.dp.⁸ For this clustering we use inverse variance weights, with the variance estimated by the delta method with second order terms as has been previous described.^{9,10}

$$var(\hat{\theta}_j) \approx \frac{\hat{\sigma}_{2,j}^2}{\hat{\beta}_{1,j}^2} + \frac{\hat{\sigma}_{1,j}^2 \hat{\beta}_{2,j}^2}{\hat{\beta}_{1,j}^4} - \frac{2\hat{\rho}\hat{\sigma}_{1,j}\hat{\sigma}_{2,j}\hat{\beta}_{2,j}}{\hat{\beta}_{1,j}^3}$$

Using this initial hard assignment to K clusters, we then perform weighted least squares (WLS) regression of $\hat{\beta}_{2,j}$ on $\hat{\beta}_{1,j}$ within each cluster with weights inversely proportional to $\hat{\sigma}_{1,j}^2 + \hat{\sigma}_{2,j}^2$.

The starting values for the EM estimation are then given by:

- π_k : the proportion of SNPs in each of the K clusters
- θ_k : the estimates slopes from WLS in each cluster
- $\beta_{1,j}$: the observed values of $\hat{\beta}_{1,j}$

3.2.2 E step

It can be shown that for the desired mixture model the expected complete data log likelihood function is given by

$$Q(\theta, \pi, \beta_1 | \theta^{(t)}, \pi^{(t)}, \beta_1^{(t)}) = \sum_{j=1}^J \sum_{k=1}^K p_{j,k} \times \left[\ln(\pi_k^{(t)}) + \ln \left\{ \varphi \left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix}; \begin{bmatrix} \beta_{1,j}^{(t)} \\ \theta_k^{(t)} \beta_{1,j}^{(t)} \end{bmatrix}, \begin{bmatrix} \hat{\sigma}_{1,j}^2 & \hat{\rho}\hat{\sigma}_{1,j}\hat{\sigma}_{2,j} \\ \hat{\rho}\hat{\sigma}_{1,j}\hat{\sigma}_{2,j} & \hat{\sigma}_{2,j}^2 \end{bmatrix} \right) \right\} \right] \quad (24)$$

where

$$p_{j,k} = \frac{\pi_k^{(t)} \varphi \left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix}; \begin{bmatrix} \beta_{1,j}^{(t)} \\ \theta_k^{(t)} \beta_{1,j}^{(t)} \end{bmatrix}, \begin{bmatrix} \hat{\sigma}_{1,j}^2 & \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} \\ \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} & \hat{\sigma}_{2,j}^2 \end{bmatrix} \right)}{\sum_k \pi_k^{(t)} \varphi \left(\begin{bmatrix} \hat{\beta}_{1,j} \\ \hat{\beta}_{2,j} \end{bmatrix}; \begin{bmatrix} \beta_{1,j}^{(t)} \\ \theta_k^{(t)} \beta_{1,j}^{(t)} \end{bmatrix}, \begin{bmatrix} \hat{\sigma}_{1,j}^2 & \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} \\ \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} & \hat{\sigma}_{2,j}^2 \end{bmatrix} \right)} \quad (25)$$

are the posterior probabilities that SNP j is a member of component k conditional on the observed data and current parameter estimates $\theta^{(t)}, \pi^{(t)}, \beta_1^{(t)}$. By design $\sum_k p_{j,k} = 1$ is maintained for each SNP j .

3.2.3 M step

It can be shown that the expected log likelihood for the E step is maximized by updated parameter estimates

$$\pi_k^{(t+1)} = \frac{1}{N} \sum_{j=1}^J p_{j,k} \quad (26)$$

$$\beta_{1,j}^{(t+1)} = \frac{\sum_{k=1}^K p_{j,k} \left[(\hat{\sigma}_{2,j}^2 - \theta_k^{(t+1)} \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j}) \hat{\beta}_{1,j} + (\theta_k^{(t+1)} \hat{\sigma}_{1,j}^2 - \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j}) \hat{\beta}_{2,j} \right]}{\sum_{k=1}^K p_{i,k} \left[\hat{\sigma}_{2,j}^2 + (\theta_k^{(t+1)})^2 \hat{\sigma}_{1,j}^2 - 2 \theta_k^{(t+1)} \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} \right]} \quad (27)$$

$$\theta_k^{(t+1)} = \frac{\sum_{j=1}^J \frac{p_{i,j} \beta_{1,j}^{(t+1)} \left[\hat{\sigma}_{1,j}^2 \hat{\beta}_{2,j} + \hat{\rho} \hat{\sigma}_{1,j} \hat{\sigma}_{2,j} (\hat{\beta}_{1,j} - \beta_{1,j}^{(t+1)}) \right]}{\hat{\sigma}_{1,j}^2 \hat{\sigma}_{2,j}^2 (1 - \hat{\rho}^2)}}{\sum_{j=1}^J \frac{p_{i,j} \hat{\sigma}_{1,j}^2 (\beta_{1,j}^{(t+1)})^2}{\hat{\sigma}_{1,j}^2 \hat{\sigma}_{2,j}^2 (1 - \hat{\rho}^2)}} \quad (28)$$

Since the estimators of $\beta_{1,j}^{(t+1)}$ and $\theta_k^{(t+1)}$ are interdependent, they are updated iteratively until convergence within each M step. This requirement is unsurprising since the same procedure is required for estimation in conventional York regression.^{2,1}

3.2.4 Convergence

After each M step, we check for convergence of the EM algorithm based on the relative improvement in the expected log likelihood function

$$\frac{Q(\theta^{(t+1)}, \pi^{(t+1)}, \beta_1^{(t+1)} | \theta^{(t)}, \pi^{(t)}, \beta_1^{(t)}) - Q(\theta^{(t)}, \pi^{(t)}, \beta_1^{(t)} | \theta^{(t)}, \pi^{(t)}, \beta_1^{(t)})}{Q(\theta^{(t)}, \pi^{(t)}, \beta_1^{(t)} | \theta^{(t)}, \pi^{(t)}, \beta_1^{(t)})} < tolerance \quad (29)$$

with an arbitrarily small tolerance value. A limit on the total number of iterations can also be applied, but in practice we find that convergence is usually fairly quick. Once the EM algorithm reached convergence, the fitted parameters $\theta^{(t+1)}, \pi^{(t+1)}, \beta_1^{(t+1)}$ are adopted as the maximum

likelihood estimates of the slopes θ_k , component proportions π_k , and marginal effect sizes on the first phenotype $\beta_{j,1}$, respectively.

3.3 Estimating residual correlation

The above estimation procedure assumes that there is an observed estimate of $\hat{\rho}$, the correlation of sampling error between the two GWAS effect sizes $\hat{\beta}_{1,j}$ and $\hat{\beta}_{2,j}$. Although this parameter isn't commonly estimated, it has previously been noted¹¹ that if $\hat{\rho}$ is assumed to be constant across genetic variants in a pair of GWAS then it can be estimated using the intercept parameter fit in genetic correlation analysis by LD score regression (LDSR)¹².

Specifically, if α_{rg} is the covariance intercept from LDSR analysis of genetic correlation between the GWAS of phenotypes 1 and 2, and if $\alpha_{h2,1}$ and $\alpha_{h2,2}$ are the univariate intercept estimates from LDSR of the same GWASs of phenotypes 1 and 2, respectively, then we estimate

$$\hat{\rho} = \frac{\alpha_{rg}}{\sqrt{\alpha_{h2,1}\alpha_{h2,2}}} \quad (29)$$

3.4 Selecting number of components

The EM algorithm estimates the desired regression mixture model for a fixed number of components K . We can then estimate the model for a range of possible values of K and use comparative model fit to evaluate the value of K that is best supported by the observed GWAS effect sizes.

Using the marginal likelihood function (Equation 23) optimized for each value of K ($\mathcal{L}_K$), we evaluate comparative model fit following the Bayesian Information Criterion (BIC)

$$BIC = (2K - 1) \ln(J) - 2 \ln(\mathcal{L}_K) \quad (29)$$

where J is the sample size (i.e. number of included SNPs) and $2K - 1$ are the model degrees of freedom from K slope parameters and $K - 1$ free component proportions π_k (i.e. due to the constraint $\sum_k \pi_k = 1$). This omits the nuisance parameters $\beta_{1,j}$ from consideration in the relative fit; since the number of nuisance parameters is constant across values of K , this amounts to a heuristic choice for the penalty term for model complexity.

Based on this criterion, the value of K that corresponds to the model with the lowest BIC is chosen as the best fitting model. Importantly, this does not mean that a model with K components fits significantly better than mixtures with other values of K . Therefore before concluding that there is evidence for distinct mechanisms in a given comparison we also strongly recommend evaluating whether the data is sufficient to reject a null hypothesis that all variants reflect a single component ($K = 1$). This hypothesis test corresponds to the test of global heterogeneity implemented in SCOUTJOY.¹³

3.5 Conclusion

Following this approach, MAGICMAP uses independent genome-wide significant variants from GWAS of a target trait (phenotype 1) and evaluates whether those loci reflect distinct mechanisms underlying the target trait and a comparator trait (phenotype 2). Under certain conditions (including a requirement that the two GWASs originate from the same population) variants that correspond to a single mechanism are expected to share the same ratio of effect sizes between the two GWAS; variants sharing this same ratio can be identified as a component within the specified regression mixture model. MAGICMAP provides estimates of the number of observed components, along with the posterior probability that each lead variant is part of each component and slope estimate that reflect the relative association of the component with the two phenotypes.

Users are encouraged to think carefully about the choice of comparisons and the interpretation of results from MAGICMAP. This description is provided for initial release, but additional information on the performance and interpretation of MAGICMAP will be forthcoming.

4 References

1. York, D., Evensen, N. M., Martínez, M. L. & De Basabe Delgado, J. Unified equations for the slope, intercept, and standard errors of the best straight line. *Am. J. Phys.* **72**, 367–375 (2004).
2. York, D. Least squares fitting of a straight line with correlated errors. *Earth Planet. Sci. Lett.* **5**, 320–324 (1969).
3. Titterton, D. M. & Halliday, A. N. On the fitting of parallel isochrons and the method of maximum likelihood. *Chem. Geol.* **26**, 183–195 (1979).
4. Fuller, W. A. *Measurement Error Models*. (Wiley, 1987). doi:10.1002/9780470316665.
5. Adcock, R. J. A Problem in Least Squares. *The Analyst* **5**, 53 (1878).
6. Dempster, A. P., Laird, N. M. & Rubin, D. B. Maximum Likelihood from Incomplete Data Via the EM Algorithm. *J. R. Stat. Soc. Ser. B Stat. Methodol.* **39**, 1–22 (1977).
7. McLachlan, G. & Peel, D. *Finite Mixture Models*. (Wiley, 2000). doi:10.1002/0471721182.
8. Song, M. & Zhong, H. Efficient weighted univariate clustering maps outstanding dysregulated genomic zones in human cancers. *Bioinformatics* **36**, 5027–5036 (2020).
9. Burgess, S. & Bowden, J. Integrating summarized data from multiple genetic variants in Mendelian randomization: bias and coverage properties of inverse-variance weighted methods. Preprint at <https://doi.org/10.48550/ARXIV.1512.04486> (2015).
10. Thomas, D. C., Lawlor, D. A. & Thompson, J. R. Re: Estimation of Bias in Nongenetic Observational Studies Using “Mendelian Triangulation” by Bautista et al. *Ann. Epidemiol.* **17**, 511–513 (2007).
11. Turley, P. *et al.* Multi-trait analysis of genome-wide association summary statistics using MTAG. *Nat. Genet.* **50**, 229–237 (2018).
12. Bulik-Sullivan, B. *et al.* An atlas of genetic correlations across human diseases and traits. *Nat. Genet.* **47**, 1236–1241 (2015).
13. Elliott, A. *et al.* Distinct and shared genetic architectures of gestational diabetes mellitus and type 2 diabetes. *Nat. Genet.* **56**, 377–382 (2024).